

Data Validation Checks

1. Quick Sanity Check

```
01_create_table.sql x 01_data_validation.sql x
superstore_db/postgres@PostgreSQL 18
No limit
Query Query History
1  /* =====
2      SuperStore Sales SQL Project
3      Data Validation
4      Author: Asiya Nizam
5      Tool: PostgreSQL (pgAdmin)
6      ===== */
7
8  /*
9  NOTE:
10 Highlight and run each query individually for the best readability of outputs.
11 */
12
13 /* =====
14 1. Quick Sanity Check
15 =====*/
16
17 SELECT *
18 FROM orders
19 LIMIT 10;
20
```

Output:

row_id	integer	order_id	character varying (50)	order_date	date	ship_date	date	ship_mode	character varying (50)	customer_id	character varying (50)	customer_name	character varying (100)	segment	character varying (50)	country_region	character varying (100)	city	character varying (100)
1	1	US-2023-103800		2023-01-03		2023-01-07		Standard Class		DP-13000		Darren Powers		Consumer		United States		Houston	
2	2	US-2023-112326		2023-01-04		2023-01-08		Standard Class		PO-19195		Phillina Ober		Home Office		United States		Naperville	
3	3	US-2023-112326		2023-01-04		2023-01-08		Standard Class		PO-19195		Phillina Ober		Home Office		United States		Naperville	
4	4	US-2023-112326		2023-01-04		2023-01-08		Standard Class		PO-19195		Phillina Ober		Home Office		United States		Naperville	
5	5	US-2023-141817		2023-01-05		2023-01-12		Standard Class		MB-18085		Mick Brown		Consumer		United States		Philadelphia	
6	6	US-2023-167199		2023-01-06		2023-01-10		Standard Class		ME-17320		Maria Etezadi		Home Office		United States		Henderson	
7	7	US-2023-167199		2023-01-06		2023-01-10		Standard Class		ME-17320		Maria Etezadi		Home Office		United States		Henderson	
8	8	US-2023-106054		2023-01-06		2023-01-07		First Class		JO-15145		Jack O'Briant		Corporate		United States		Athens	
9	9	US-2023-167199		2023-01-06		2023-01-10		Standard Class		ME-17320		Maria Etezadi		Home Office		United States		Henderson	
10	10	US-2023-167199		2023-01-06		2023-01-10		Standard Class		ME-17320		Maria Etezadi		Home Office		United States		Henderson	

2. Data Type Inspection

```

21  /* =====
22      2. Data Type Inspection
23      =====*/
24
25  SELECT  column_name,
26          data_type
27  FROM    information_schema.columns
28  WHERE   table_name = 'orders'
29  ORDER BY data_type;
30

```

Output:

	column_name 	data_type 
1	customer_name	character varying
2	postal_code	character varying
3	region	character varying
4	product_id	character varying
5	category	character varying
6	order_id	character varying
7	sub_category	character varying
8	ship_mode	character varying
9	customer_id	character varying
10	segment	character varying
11	country_region	character varying
12	city	character varying
13	state_province	character varying
14	order_date	date
15	ship_date	date
16	row_id	integer
17	quantity	integer
18	discount	numeric
19	profit	numeric
20	sales	numeric
21	product_name	text

3. Row Count Check

```
31  /* =====
32      3. Row Count Check
33      =====*/
34
35  SELECT COUNT(*) AS total_rows
36  FROM ORDERS;
37
```

Output:

	total_rows bigint 
1	10194

4. Duplicate Check (Row Id Level)

```
38  /* =====
39      4. Duplicate Check (Row_id Level)
40      =====*/
41
42  SELECT row_id,
43         COUNT(*) AS occurences
44  FROM orders
45  GROUP BY row_id
46  HAVING COUNT(*) > 1;
47
```

Output:

	row_id integer 	occurences bigint 

5. Orders Spanning Multiple Rows

```

48  /* =====
49      5. Orders Spanning Multiple Rows
50      (expected – one order can have many products)
51      =====*/
52
53  SELECT  order_id,
54          COUNT(*) AS row_count
55  FROM    orders
56  GROUP BY order_id
57  HAVING  COUNT(*) > 1
58  ORDER BY row_count DESC
59  LIMIT 10;
60

```

Output:

	order_id character varying (50) 	row_count bigint 
1	US-2026-100111	14
2	US-2026-157987	12
3	US-2025-108504	11
4	US-2025-165330	11
5	US-2024-126977	10
6	US-2024-131338	10
7	US-2025-105732	10
8	US-2026-140949	9
9	US-2026-117457	9
10	US-2024-132626	9

6. Missing Values Check (Data Quality)

```

61  /* =====
62      6. Missing Values Check (Data Quality)
63      =====*/
64
65  SELECT
66      COUNT(*) FILTER(WHERE sales IS NULL) AS missing_sales,
67      COUNT(*) FILTER(WHERE profit IS NULL) AS missing_profit,
68      COUNT(*) FILTER(WHERE order_date IS NULL) AS missing_order_date,
69      COUNT(*) FILTER(WHERE customer_name IS NULL) AS missing_customer
70  FROM orders;
71

```

Output:

	missing_sales bigint	missing_profit bigint	missing_order_date bigint	missing_customer bigint
1	0	0	0	0

7. Invalid Sales

```
72  /* =====
73      7. Invalid Sales
74      =====*/
75
76  SELECT COUNT(*) AS negative_sales
77  FROM orders
78  WHERE sales < 0;
79
80  SELECT COUNT(*) AS zero_sales
81  FROM orders
82  WHERE sales = 0;
83
```

Outputs:

	negative_sales bigint
1	0

	zero_sales bigint
1	0

8. Invalid Quantity

```
84  /* =====
85      8. Invalid Quantity
86      =====*/
87
88  SELECT COUNT(*) AS invalid_quantity
89  FROM orders
90  WHERE quantity <= 0;
91
```

Output:

	invalid_quantity bigint	
1	0	

9. Discount Range Check

```
92  /* =====
93      9. Discount Range Check
94      =====*/
95
96  SELECT COUNT(*) AS discount_out_of_range
97  FROM orders
98  WHERE discount < 0 OR discount > 1;
99
```

Output:

	discount_out_of_range bigint	
1	0	

10. Date Range Check

```
100 /* =====
101     10. Date Range Check
102     =====*/
103
104  SELECT MIN(order_date) AS earliest_order_date,
105         MAX(order_date) AS latest_order_date
106  FROM orders;
107
```

Output:

	earliest_order_date date	latest_order_date date
1	2023-01-03	2026-12-30

11. Ship Date Before Order Date Check

```

108  /* =====
109      11. Ship Date Before Order Date Check
110      =====*/
111
112  SELECT COUNT(*) AS ship_before_order
113  FROM orders
114  WHERE ship_date < order_date;
115

```

Output:

	ship_before_order bigint
1	0

12. Future Order Date Check

```

116  /* =====
117      12. Future Order Date Check
118      =====*/
119
120  SELECT COUNT(*) AS future_order_dates
121  FROM orders
122  WHERE order_date > CURRENT_DATE;
123

```

Output:

	future_order_dates bigint
1	2465

13. Distinct Category Check

```

124  /* =====
125      13. Distinct Category Check
126      =====*/
127
128  SELECT DISTINCT ship_mode FROM orders ORDER BY ship_mode;
129  SELECT DISTINCT region FROM orders ORDER BY region;
130  SELECT DISTINCT country_region FROM orders ORDER BY country_region;
131  SELECT DISTINCT category FROM orders ORDER BY category;
132  SELECT DISTINCT sub_category FROM orders ORDER BY sub_category;
133

```

Outputs:

	ship_mode character varying (50) 
1	First Class
2	Same Day
3	Second Class
4	Standard Class

	region character varying (50) 
1	Central
2	East
3	South
4	West

	country_region character varying (100) 
1	Canada
2	United States

	category character varying (50) 
1	Furniture
2	Office Supplies
3	Technology

	sub_category character varying (50) 🔒
1	Accessories
2	Appliances
3	Art
4	Binders
5	Bookcases
6	Chairs
7	Copiers
8	Envelopes
9	Fasteners
10	Furnishings
11	Labels
12	Machines
13	Paper
14	Phones
15	Storage
16	Supplies
17	Tables

14. Business KPI Sanity Check

	total_sales numeric 🔒	total_profit numeric 🔒	unique_orders bigint 🔒	unique_customers bigint 🔒
1	2326534.52	292297.57	5111	804